

WEEKLY JOURNAL DIGEST

Weekly journal digest for 2026-06-15 to 2026-06-21

Curated articles in political communication and computational social science.

SUMMARY

COMAP Journal Bot reviewed 319 collected candidates for the 2026-06-15 to 2026-06-21 digest window. After manual relevance filtering, the curated digest keeps 38 papers: 15 new this week, 23 previous-week catch-up items, and 0 late additions.

This early-cycle brief is especially strong on the institutional and behavioral consequences of mediated decision-making. The new papers connect persuasive targeting to attitude-relevant beliefs, corrosive internet-news frames to democratic institutions, residential design to grassroots politics, and behavioral nudges to autonomous AI agents. A second cluster focuses on inequality and governance: climate-disaster vulnerability, violence against women, antimicrobial stewardship, agricultural multi-hazard exposure, and uncertainty in historical wealth evidence.

- Political communication leads the week through persuasion, institutional threat framing, and the governance of science agencies.
- AI-for-CSS appears in work on nudge-sensitive agents and commentary on video games as testbeds for advancing AI.
- Policy-facing research spans climate risk, antimicrobial markets, violence against women, wildfire conditions, and the effects of war on wildlife behavior.

HIGHLIGHTS

Beyond Belief Change: The Persuasive Returns of Targeting Attitude-Relevant Beliefs

AMERICAN POLITICAL SCIENCE REVIEW · 2026-06-17

YAMIL RICARDO VELEZ; PATRICK LIU; SCOTT CLIFFORD

Why it matters · This meta-reanalysis sharpens how political persuasion is measured by separating attitude-relevant beliefs from generic belief change, making it useful for message targeting and experimental design.

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Vernacular Architecture and Grassroots Urban Politics: How Politics Is Embedded in Residential Design

AMERICAN POLITICAL SCIENCE REVIEW · 2026-06-17

PAIGE BOLLEN; NOAH L. NATHAN

Why it matters · The paper connects residential design to grassroots urban politics, showing how political organization can be embedded in the built environment rather than only in formal institutions.

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Stewardship without walls: taming the unregulated antibiotic economy to contain antimicrobial resistance in Ethiopia

NATURE COMMUNICATIONS · 2026-06-18

Balew Arega; Asnake Agunie

Why it matters · A clear governance contribution on antimicrobial resistance: it treats informal antibiotic markets as an institutional stewardship problem rather than only a clinical one.

[Open article →](#)

Inequality in human development amplifies climate-related disaster risk

NATURE COMMUNICATIONS · 2026-06-17

Khalil Teber; Sebastian Sippel; Melanie Krause; Jakob Zscheischler; Miguel D. Mahecha

Why it matters · This paper directly links inequality in human development to climate-disaster vulnerability, offering a policy-relevant account of how social capacity shapes environmental risk.

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Democratic Institutions Under Threat: Corrosive Frames, Internet News, and the U.S. Supreme Court

POLITICAL COMMUNICATION · 2026-06-17

Justin Wedeking; Michael A. Zilis

Why it matters · A strong political-communication fit: it examines how corrosive frames and internet news can weaken support for democratic institutions, using the U.S. Supreme Court as the focal case.

[Open article →](#)

Is the elimination of violence against women a realistic goal for the near future?

NATURE COMMUNICATIONS · 2026-06-16

Enrique Gracia

Why it matters · This perspective assesses whether eliminating violence against women is feasible on the current policy trajectory and identifies where institutional targets are falling short.

[Open article →](#)

AI agents are sensitive to nudges

PROCEEDINGS OF THE NATIONAL ACADEMY OF SCIENCES · 2026-06-15

Manuel Cherep; Pattie Maes; Nikhil Singh

Why it matters · A concrete AI-governance result: autonomous LLM agents respond to behavioral nudges, raising practical concerns for delegated decisions and tool use.

[Open article →](#)

Accounting for uncertainty and bias in archaeological and historical evidence on wealth inequality

PROCEEDINGS OF THE NATIONAL ACADEMY OF SCIENCES · 2026-06-15

Mattia Fochesato; Samuel Bowles

Why it matters · The paper develops methods for handling uncertainty and bias in long-run wealth-inequality evidence, with value for historical inference and comparative institutional research.

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Previous Week Catch-Up

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COLLECTION SNAPSHOT		
TOTAL COLLECTED CANDIDATES 319	NEW THIS WEEK CANDIDATES 157	PREVIOUS WEEK CATCH-UP CANDIDATES 162
LATE ADDITIONS CANDIDATES 0	CURATED DIGEST BELOW 15 new, 23 catch-up, 0 late additions	NOTE The full collected set above is before manual relevance filtering, while the HTML email body, attached full HTML artifact, and attached PDF reflect the curated digest.

FULL CURATED DIGEST

New This Week

Science Advances

Heatwaves enable wildfire activity in the western United States

2026-06-19 · DOI 10.1126/SCIADV.AEA1277

Dmitri A. Kalashnikov; John T. Abatzoglou; Emily L. Williams; Cong Yin; Madhulika Gurazada; Mukesh Kumar; Ashwin P. Thomas; Precious E. Ebiendele

School of Engineering, University of California, Merced, Merced, CA, USA.; Sierra Nevada Research Institute, University of California, Merced, Merced, CA, USA.

Abstract. While overall impacts of heatwaves have been extensively studied, the connection between heatwaves and wildfire activity remains relatively underexplored. We analyze links between heatwaves and both wildfire occurrence and growth across the western United States (WUS) and find that 42% of burned area during 2001-2024 occurred during and immediately following heatwaves. Heatwaves facilitate significant increases in daily burned area through meteorological and fuel flammability conditions that promote new ignitions and exacerbate ongoing fire activity, with effects persisting post-heatwave in most regions. In addition, heatwaves co-occur with increased cloud-to-ground lightning that can potentially increase ignitions. Last, we observe a 2.5-fold increase in burned area in WUS forests since 2001, with ~64% of this increase coinciding with heatwaves, but without corresponding increases in nonforests. The growing influence of heatwaves in shaping burned area in WUS forests has important implications for fire management and public health and can improve predictions of wildfire risk.

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Nature Communications

Multi-hazard exposure and vulnerability drive rising agricultural risk in global breadbasket regions

2026-06-18 · DOI 10.1038/S41467-026-74617-5

Hossein Tabari; Parisa Hosseinzadehtalaei

Abstract. Abstract Climate extremes increasingly threaten global food production, yet global multi-hazard agricultural risk assessments that jointly consider multiple hazards, crop exposure and socioeconomic vulnerability are still missing. This gap limits our ability to identify where climate shocks to croplands are most likely to translate into high agricultural risk. Here we present a global agricultural risk assessment that quantifies the exposure of croplands and ten major crop groups to heatwaves, droughts, floods and compound hot-dry events, and combines this with a vulnerability index constructed from socio-economic and institutional indicators using latent factor modelling. Using ensembles of climate and hydrological models under multiple future scenarios, we show widespread increases in multi-hazard exposure, with the strongest intensification across South Asia, Sub-Saharan Africa and major U.S. and European breadbaskets. A significant relationship between vulnerability and cropland exposure emerges in future scenarios but is absent historically, indicating a growing role for vulnerability in shaping future agricultural risk. These results demonstrate that reliably identifying future agricultural risk hotspots requires jointly accounting for multi-hazard exposure and socioeconomic vulnerability, so that adaptation efforts can be targeted to regions where intensifying extremes coincide with high vulnerability.

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Stewardship without walls: taming the unregulated antibiotic economy to contain antimicrobial resistance in Ethiopia

2026-06-18 · DOI 10.1038/S41467-026-74671-Z

Balew Arega; Asnake Agunie

Abstract. Ethiopia's unregulated private antibiotic economy spans private hospitals, clinics, pharmacies, drug shops and informal vendors. It drives resistance through three forces: intergenerational self-medication (reservoir habit), a satisfaction imperative linking sales to customer retention, and a corner drugstore calculus prioritising immediate access over cheaper but slow public care. Non-prescribed dispensing becomes the norm and selects resistant bacteria. We propose a six-pillar framework: tiered accreditation, point-of-service education, supply chain engagement, mystery client support, technology enabled consultation and operational research. This stewardship without walls is an urgent priority for Ethiopia and similar settings. Arega and Agunie analyze Ethiopia's unregulated private antibiotic economy and propose a six-pillar framework of stewardship without walls to contain antimicrobial resistance in low-income settings.

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Inequality in human development amplifies climate-related disaster risk

2026-06-17 · DOI 10.1038/S41467-026-73873-9

Khalil Teber; Sebastian Sippel; Melanie Krause; Jakob Zscheischler; Miguel D. Mahecha

Abstract. Abstract The impacts of climate-related disasters are shaped by the interaction between hazard intensity, exposure, and vulnerability. However, the influence of hazard intensity and within-country inequality on impact magnitudes remains poorly quantified. Here, we present a global multi-hazard study of over 7000 climate-related disasters reported by the Emergency Events Database from 1990 to 2020. Using subnational indicators, we show that human development drives major shifts in global exposure and impact patterns, with societal vulnerability outweighing hazard intensity in shaping impacts. Despite a declining share of global exposure over the past three decades, regions with low subnational Human Development Index scores experience disproportionately higher human losses across most disaster types. For instance, individuals in these regions face an 8.2-fold higher risk of fatality associated with storms (95% confidence interval: 2.16-23.06) compared to those in very high human development regions. Our findings also indicate that within-country inequality in human development exacerbates disaster risk in regions with low and medium levels of human development. These results underscore the critical role of human development in managing disaster risks and highlight the link between socioeconomic conditions and vulnerability to climate-related hazards.

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Is the elimination of violence against women a realistic goal for the near future?

2026-06-16 · DOI 10.1038/S41467-026-74656-Y

Enrique Gracia

Abstract. As 2030 approaches, achieving Sustainable Development Goal Target 5.2, which aims to eliminate all forms of violence against women, appears increasingly unrealistic. Using the European Union as a case study, this Perspective argues that prevalence remains high, progress has stagnated, levels can be disproportionately high in some of the most socioeconomically advanced and gender-equal countries, and prevention efforts seem unable to reach younger generations. Together, these patterns suggest that prevention and response efforts, as implemented to date, have not produced sustained population-level reductions. Achieving measurable declines will require scaling evidence-based prevention with sustained political will and stronger accountability. This Perspective argues that violence against women remains persistently high, in some of the most socioeconomically advanced and gender-equal countries as well. Progress has stagnated and prevention efforts seem unable to reach younger generations.

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Proceedings of the National Academy of Sciences

Video games help push the boundaries of AI

2026-06-18 · DOI 10.1073/PNAS.2618819123

Stephen Ornes

Abstract. In one recent project that used video games to explore artificial intelligence, computer scientists introduced 1,000 AI-driven agents into the game Minecraft and watched as an AI civilization emerged. It included artists, chefs, and explorers. Image credit: Shutterstock/ mkfilm. A few years ago, a group of computer scientists set 25 players loose in Smallville, a custom-made virtual "sandbox world"—an open-ended video game like Minecraft that allows players to freely explore and interact with each other. The catch: The players in Smallville were all virtual, each driven by its own large language model, or LLM. The goal, says Joon Sung Park, who led the project while finishing his PhD at Stanford University in California, was to see what kinds of human-like behaviors evolved. To get the players started, Park and his colleagues wrote a paragraph describing the identity of each. For example, for one they wrote, "John Lin is a pharmacy shopkeeper at the Willow Market and Pharmacy who loves to help people." What happened next was up to the LLMs. In the spring of 2023, the researchers reported on what unfolded over two game days in Smallville (1). The game's AI populace had been surprisingly busy with familiar-seeming tasks. "They would wake up in the mornings, go to work, and do all kinds of things," Park says. When one agent was instructed (by a researcher on Park's team) to throw a Valentine's Day party with no further instructions, it issued invitations and began party preparations. Some Smallvillians began asking dates to the party and making plans to get there on time. When one agent told another that it planned to run for mayor, the campaign quickly became a hot topic in Smallville as agents discussed how they planned to vote. The sandbox environment gave Park a way to explore human behavior using LLMs; his study is one of the latest in a long history of using games to push computer technology forward. Some scientists develop games to explain difficult concepts, or even to crowdsource new ideas in a field (2). "When it comes to developing artificial intelligence, games were there from the beginning," says Julian Togelius, a computer scientist and video game researcher at New York University who explores the intersection of AI and video games. Games like chess and Go, he says, gave early researchers a way to test the capabilities of new AI tools—and remain valuable touchstones for new insights (3).

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AI agents are sensitive to nudges

2026-06-15 · DOI 10.1073/PNAS.2537030123

Manuel Cherep; Pattie Maes; Nikhil Singh

Department of Computer Science, Dartmouth College; Media Lab, Massachusetts Institute of Technology

Abstract. Large language models (LLMs) are increasingly deployed as autonomous agents that make choices and use tools on behalf of users. Yet, we have limited evidence about how their decisions are shaped by their environment. We adapt a human decision-making task to test leading LLMs under four forms of choice architecture: defaults, suggestions, information highlighting, and "optimal" nudges derived from a resource-rational model of human choice. We treat human behavior as a baseline for predictable sensitivity to such interventions. Across models and prompting strategies, LLMs often depart substantially from this baseline. They sometimes pay excessive costs to acquire information, sometimes ignore available information, and, most crucially, are far more responsive to nudges than humans, such that weak cues that slightly shift human behavior have larger effects on model choices, toward both better and worse payoff outcomes. Chain-of-thought prompting and in-context human data do not reliably stabilize behavior. Recent reasoning-optimized LLMs can, in some configurations, restore more human-level sensitivity to nudges, but do so inconsistently and at substantial computational cost. These results point to an important and largely neglected safety concern: LLM agents can be behaviorally brittle under subtle changes in choice architecture, even in the absence of adversarial settings.

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Accounting for uncertainty and bias in archaeological and historical evidence on wealth inequality

2026-06-15 · DOI 10.1073/PNAS.2514309123

Mattia Fochesato; Samuel Bowles

CORE Economics, University College London; Centre for Economic Policy Research; Department of Social and Political Sciences, Dondena, Centre for Research on Geography, Resources, Environment, Energy & Networks, and Bocconi Institute for Data Science and Analytics-Bocconi University; Santa Fe Institute

Abstract. The archaeological and historical record allows us to study the patterns of economic inequalities associated with sharply contrasting institutions such as communal property in societies without states, private property among small holders, and slavery, in economies relying on radically different technologies, based for example on human energy alone, other animal power, and carbon-based power. But the price of using prehistoric and historical data to expand the range of institutions and technologies under study is a substantial and typically unknown level of uncertainty and unrepresentativeness in the resulting estimates. We provide methods and code that quantify the degree of uncertainty, an approach that we term BRIDGE (Bayesian-Resampling and Informed Priors with Data-driven Gini Estimation). We transform raw wealth data from 431 sites and dates into probability distributions representing likely levels of inequality that (insofar as possible) are representative of the underlying population (e.g., rectifying biases arising from small or nonrandom samples and the frequent absence of data on those entirely without wealth) and are comparable across differing asset types (e.g., burial goods, dwelling area, storage area, land cultivated), ownership units (individual or household), and scale (from villages to nations). These distributions are based on the raw data along with independent information that allows us to infer biases and levels of uncertainty. We also account for the propagation of uncertainty through every stage of the data generation and estimation process. We find that widely used methods (e.g., relying solely on conventional bootstrapping) provide misleading and for the most part substantially underestimated measures of uncertainty.

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Oxytocin modulates the neurocomputational mechanisms engaged in learning rank relationships in social networks

2026-06-15 · DOI 10.1073/PNAS.2606871123

Jiawei Liu; Chen Qu; Remi Philippe; Siying Li; Edmund Derrington; Jean-Claude Dreher

Faculty of Education, Northeast Normal University; Key Laboratory of Brain, Cognition and Education Sciences (South China Normal University), Ministry of Education; Laboratory of Neuroeconomics, Institut des Sciences Cognitives Marc Jeannerod; School of Psychology, Center for Studies of Psychological Application, and Guangdong Key Laboratory of Mental Health and Cognitive Science, South China Normal University; Universite Claude Bernard Lyon 1

Abstract. Indirect evidence from preclinical and neuropharmacological studies suggests that oxytocin may attenuate social dominance relationships and modulate social memory. In humans, oxytocin may also modulate self- versus other-oriented reward learning and in-group/out-group decisions differently. Although the neural bases of learning social hierarchy by observation have been identified for linear hierarchies (a 1 2 n), the brain mechanisms engaged in learning more complex rank relationships (i.e.) in social networks remain unknown. Here, we investigated the modulatory role of oxytocin on the neurocomputational mechanisms engaged when learning ranks in social networks and when making transitive inferences based on social memory retrieval. Participants learned rank relationships between members of two social networks, one in which they were embedded and the other not. During a subsequent test phase, they inferred rank relationships between pairs of members not encountered before. During training, oxytocin (vs. placebo) improved choice accuracy regarding rank relationship between pairs of members for other-oriented network and increased activity of the ventromedial prefrontal cortex. During the test phase, oxytocin modulated the balance between self/other social memory, boosting performance and amygdala responses for memory retrieval concerning social networks in which participants were not embedded. These findings demonstrate that oxytocin modulates the brain computations needed to learn and retrieve rank relationships in social networks in a manner that depends upon reference-frame.

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Science

Changes in wildlife activity patterns in response to war in Ukraine

2026-06-18 · DOI 10.1126/SCIENCE.AED1493

Svitlana Kudrenko; Richard Bischof; Andreas Zedrosser; Nuria Selva; Kateryna Korepanova; Denys Vyshnevskiy; Martin Gahbauer; Serhii Obrizan; Serhii Domashevskiy; Oleksandr Borsuk; Alona Varukha; Marco Heurich

Bavarian Forest National Park, Grafenau, Germany.; Chornobyl Radiation and Ecological Biosphere Reserve, Ivankiv, Kyiv region, Ukraine.; Departamento de Ciencias Integradas, Facultad de Ciencias Experimentales, Centro de Estudios Avanzados en Fisica, Matematicas y Computacion, Universidad de Huelva, Huelva, Spain.; Department for Integrated Biology, Institute of Wildlife Biology and Game Management, University for Natural Resources and Life Sciences, Vienna, Austria.; Estacion Biologica de Donana Consejo Superior de Investigaciones Cientificas, Sevilla, Spain.; Faculty of Environment and Natural Resources, University of Freiburg, Freiburg im Breisgau, Germany.; Faculty of Environmental Sciences and Natural Resource Management, Norwegian University of Life Sciences, As, Norway.; Faculty of Technology, Natural Sciences and Maritime Sciences, Department of Natural Sciences and Environmental Health, University of South-Eastern Norway, B, Norway.; et al. (13 total)

Abstract. Conflict zones are inherently hazardous and inaccessible for researchers, which results in a knowledge gap about the immediate effects of armed conflicts on the environment, particularly wildlife. We used camera-trap detections to investigate the impact of armed conflict on wildlife activity patterns before, during, and after the Russian occupation of the Chornobyl Exclusion Zone (Ukraine) in 2022 and compared it to the same period in 2021. Mammal species responded to armed conflict through immediate behavioral adjustments, including reduced activity during night and on dates when armed-conflict activities intensified. Our results provide insight into wildlife's behavioral responses to armed conflict in real time and underscore the potential of camera trapping to quantify the ecological effects of war.

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New list of political jobs at science agencies called 'weird and capricious'

2026-06-18 · DOI 10.1126/SCIENCE.AEJ7945

Jeffrey Mervis

Abstract. NIH programs and leadership are hit hard; other science agencies are barely mentioned

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American Political Science Review

Beyond Belief Change: The Persuasive Returns of Targeting Attitude-Relevant Beliefs

2026-06-17 · DOI 10.1017/S0003055426101622

YAMIL RICARDO VELEZ; PATRICK LIU; SCOTT CLIFFORD

Columbia University; Texas A&M University

Abstract. A persistent puzzle in the study of public opinion is why political information often produces minimal attitude change despite reliably influencing beliefs. We argue that this duality reflects belief relevance—the extent to which specific beliefs bear on attitudes. Using conversations with large language models (LLMs), we elicit deeply held issue attitudes and the “focal beliefs” people use to justify those attitudes. We then randomly assign participants to receive an LLM-generated counterargument targeting either their focal belief, an issue-relevant but unmentioned belief (“distal belief”), or a placebo. In experiments with two large online convenience samples, counterarguments targeting the aforementioned beliefs successfully decrease belief strength, with effects persisting after 10 days. More importantly, focal belief counterarguments produce larger and more durable attitude change than distal counterarguments. These findings suggest that political information can successfully shift political attitudes and provide new evidence for the role of belief relevance in persuasion.

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Vernacular Architecture and Grassroots Urban Politics: How Politics Is Embedded in Residential Design

2026-06-17 · DOI 10.1017/S0003055426101592

PAIGE BOLLEN; NOAH L. NATHAN

Massachusetts Institute of Technology; The Ohio State University

Abstract. The physical structures in which urban life occurs are an underappreciated determinant of how grassroots urban politics unfolds. In many rapidly growing cities, housing scarcity forces residents into multifamily buildings that create daily exposure to neighbors. We argue that these exposures affect political behavior by shaping residents' access to political information and capacity for collective action. We focus on the informal, vernacular architecture of West Africa's dominant urban housing form—the compound house. Compound house residents in urban Ghana participate more in politics than similar residents of other housing types. Leveraging an original survey, including novel measures of tenants' spatial network centrality within their residential buildings, we suggest that key mechanisms for this relationship emerge from the effects of architectural design on visibility and social ties among co-tenants. Ultimately, built environments must be studied alongside demographic environments to best understand contextual effects on political behavior.

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Political Communication

Democratic Institutions Under Threat: Corrosive Frames, Internet News, and the U.S. Supreme Court

2026-06-17 · DOI 10.1080/10584609.2026.2682952

Justin Wedeking; Michael A. Zilis

University of Kentucky

Abstract. Are democratic institutions under threat in the United States? We examine an understudied democratic bedrock: the judiciary and, in particular, the U.S. Supreme Court. We draw on data from "SCOTUSapp," a tool that provides a database of 100,000+ news articles across thousands of unexamined sources. We find a new media environment where the Court receives attention from thousands of ideologically fragmented outlets. Additionally, some framings of the Court are potentially corrosive. We then use two experiments to examine the effects of these corrosive news frames. The first experiment (October 2021) provides evidence that frames portraying the Court as embroiled in secrecy, hypocrisy, illegitimacy, and politics cause a decrease in support for judicial independence. These frames are influential among Republicans and Democrats. The second experiment (May 2023) follows *Dobbs*, which severely damaged judicial faith among Democrats, and shows that corrosive frames lead to a corresponding decline only among strong Republicans.

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Nature Human Behaviour

The interaction of meaning similarity and confusability explains regularity in form-meaning mappings at and below the word level

2026-06-15 · DOI 10.1038/S41562-026-02488-3

Thomas Brochhagen; Xixian Liao; Jamie D. Wright; Carmen Saldana

Abstract. Languages exhibit striking regularities in how meanings are mapped to word forms, yet analogous patterns at the subword level remain under-explored. This study presents a large-scale cross-linguistic analysis of regularity at and below the word level, drawing on data from over 1,900 languages. Here we find that while the co-expression of meanings at both levels

is highly systematic, the meanings recurrently involved in the two levels differ. Nevertheless, regularity at and below the word level is explained by the same underlying principle: a tension between pressure for lexical compression and lexical differentiation. The former favours reusing forms for similar meanings to ease learning and processing. The latter favours the use of distinct forms to prevent ambiguity. These findings offer a unified account of lexical organization across the world's languages, with subword-level form reuse emerging as a principled compromise when full word-level reuse risks miscommunication. Languages exhibit striking regularities in how meanings are mapped to word forms, yet analogous patterns at the subword level remain under-explored. This study fills the gap with a large-scale cross-linguistic analysis of regularity at and below the word level, drawing on data from over 1,900 languages.

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Previous Week Catch-Up

Communication Research

Co-Opetition in Crowdsourcing Challenges: Comment Network Dynamics and Relational Homophily Among Contestants in Crowdsourcing Challenge Communities

2026-06-13 · DOI 10.1177/00936502261458941

Yiqi Li; Peter Monge; Janet Fulk

Syracuse University, NY, USA; University of Southern California, Los Angeles, CA, USA

Abstract. Situated in a creative crowdsourcing challenge organizational community, this research examines how different types of homophily affect network formation among ideators, the challenge innovators. Longitudinal analysis of comment network events among ideators was conducted with relational event models. Status homophily was examined based on ideators' self-reported public profiles. Findings show that ideators' location by country significantly influences relational events, but occupational homophily did not. Value homophily was examined based on the content similarity of ideators' idea submissions. Results indicate that ideator's choices of platform content categories, those that are designed by the challenge management, significantly influenced ideators' communication patterns. Specifically, long-term needs were a salient homophily factor among platform categories, whereas short-term needs were heterophilous factors. Finally, a multidimensional homophily measurement using latent class analysis on idea-category-submission patterns predicted the occurrences of communication relational events and achieved an improved model fit compared to modeling all platform categories together.

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When the Algorithm Understands You (or Doesn't): LGBTQ+ Youth's Identity-Related Interactions with Algorithmic Media

2026-06-13 · DOI 10.1177/00936502261459750

Y. Anthony Chen; Samuel Hardman Taylor

Stanford University, CA, USA; University of California Irvine, Stanford, USA; University of Illinois Chicago, USA

Abstract. During their daily media use, youth interact with algorithms. Drawing on the algorithm responsiveness process framework, this paper examines the co-construction of identity between youth and algorithmic media, focusing on perceived algorithm responsiveness (PAR), perceived algorithm insensitivity (PAI), and algorithm training as sociotechnical processes that structure how LGBTQ+ youth perceive themselves. Study 1 (N = 335), using six weekly diaries among LGBTQ+ youth, showed that affirming LGBTQ+ representations curated by algorithms predicted higher PAR and lower PAI, especially when less algorithm training was needed. In turn, higher PAR and, unexpectedly, higher PAI predicted greater identity pride. Study

2 (N = 664), a cross-sectional survey, found that LGBTQ+ youth reported lower PAR than non-LGBTQ+ youth. An indirect effect of algorithm training on self-concept clarity via PAR emerged for non-LGBTQ+ participants but not for LGBTQ+ participants. These findings highlight social identity in algorithmic media as a relational, dynamic process, and LGBTQ+ youth can shape content recommendations for identity-related needs.

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Two Forms of Media Distrust: Cynicism, Skepticism, and Online News Exposure

2026-06-10 · DOI 10.1177/00936502261458955

Lukas P. Otto; Frank Mangold; Michaela Maier

GESIS - Leibniz Institute for the Social Sciences, Cologne, Germany; RPTU Kaiserslautern-Landau, Institute for Communication Psychology and Media Education, Germany

Abstract. The relationship between trust in the news media and news use has been studied extensively, yet findings remain inconsistent. We argue that a key source of this inconsistency is the conflation of two distinct forms of media distrust: skepticism and cynicism. Additionally, prior research largely relies on self-reported news use, which is prone to bias. To address both issues, we link survey data with web-tracking data from 739 participants during the 2021 German federal election. We show that skeptics and cynics differ in personality traits, political orientations, and news-use motivations, and - crucially - in their actual news consumption behavior. While skepticism is associated with broader and more intense news exposure, cynicism is linked to lower exposure, particularly to established journalistic mainstream media. These results demonstrate that different forms of media distrust are associated with distinct patterns of news exposure, underscoring the importance of conceptual and behavioral precision in media trust research.

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Nature Communications

A probabilistic forecasting framework for neighbourhood-level disaggregation of electric vehicle adoption scenarios

2026-06-13 · DOI 10.1038/S41467-026-74155-0

Isaac Flower; Furong Li; Julian Padgett

Abstract. Abstract The rapid growth of electric vehicle (EV) adoption presents significant challenges for electricity networks, particularly at the low-voltage level, where clustered neighbourhood demand risks overloading infrastructure. Existing scenario-based planning approaches typically assume uniform EV uptake across neighbourhoods within a region, failing to capture the heterogeneity in historical EV registration data. They provide limited uncertainty quantification, despite the difficulty of predicting future adoption at fine spatial scales. This paper introduces a Gaussian process (GP)-based forecasting framework that combines granular historical EV registration data with top-down regional scenarios to generate probabilistic neighbourhood-level forecasts. The GP captures how local adoption deviates from regional trends, encoded in the GP's mean function, ensuring consistency with broader scenarios while accounting for local variation and uncertainty. We validate the framework using ten representative local authority districts in England and Wales, covering 1,294 neighbourhoods. The framework demonstrates improved performance compared to baseline methods (scaled scenario, logistic growth, linear extrapolation) in normalised mean absolute error, with statistically significant improvements at horizons of three years and beyond. It also delivers well-calibrated prediction intervals, providing reliable uncertainty estimates. This framework offers a practical tool for network operators, policymakers, and planners to support targeted decision-making and investment.

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Steering open-source AI to accelerate the sustainable development goals

2026-06-11 · DOI 10.1038/S41467-026-73866-8

Min Chen; Kai Wu; Prajal Pradhan; Cameron Allen; Stefano Nativi; Klaus Hubacek; Alexey Voinov; Felix Creutzig; Tatiana Filatova; Niklas Boers; Michael Meadows; Peilong Ma; et al. (21 total)

Abstract. While the artificial intelligence (AI) revolution is advancing rapidly, the open-source paradigm offers key pathways and potential risks for accelerating progress towards the Sustainable Development Goals and beyond. This comment introduces four governance actions that consider how sustainability, evaluation, safety, and cooperation can be integrated into the transformation of open-source AI, thereby reducing uncertainties and challenges posed by open-source AI for sustainable global prosperity. This comment introduces four governance actions that consider how sustainability, evaluation, safety, and cooperation can be integrated into the transformation of open-source AI, thereby reducing uncertainties and challenges posed by open-source AI for sustainable global prosperity.

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Navigating adoption barriers for microbial proteins in future food

2026-06-10 · DOI 10.1038/S41467-026-73987-0

Rima Gnaim; Hakimi Kassim; Lisa Neidhardt; Thomas Gassler; Rodrigo Ledesma-Amaro

Abstract. Abstract Microbial biomass fermentation, in which microbes are cultivated to produce nutrient-dense biomass, offers a scalable route to sustainable protein production with low land, water, and greenhouse gas footprints. However, its shift from a speciality to a mainstream food continues to be difficult. Here, we move beyond technological overviews and propose a three-phase adoption framework: novelty barrier, early trust-building, and mainstream normalisation. This framework organises techno-economic, regulatory, and infrastructural barriers into a single trajectory. We trace single-cell proteins' rise, decline and resurgence, map engineering and policy enablers by phase, and outline levers to move microbial proteins into resilient food systems.

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Nature Computational Science

Transferable human mobility network reconstruction with neuroGravity

2026-06-12 · DOI 10.1038/S43588-026-01003-Y

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Abstract. Accurate modeling of human mobility is critical for tackling urban planning and public health challenges. In undeveloped regions, the absence of comprehensive travel surveys necessitates reconstructing mobility networks from publicly available data. Here we develop neuroGravity, a physics-informed deep learning model that reliably reconstructs mobility flows from limited observations and transfers to unobserved cities. Using only urban facility and population distributions, we find that neuroGravity's regional representations strongly correlate with socioeconomic and livability status, offering scalable proxies for costly surveys. Furthermore, we uncover that spatial income segregation has a key role in model transferability: mobility networks are most reliably reconstructed when target cities share similar segregation levels with the source. We design an index to quantify this segregation and accurately predict transferability. Finally, we generate mobility flow proxies for over 1,200 cities worldwide, highlighting neuroGravity's potential to mitigate critical data shortages in resource-limited, underdeveloped areas.

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Nature Human Behaviour

Social determinants of health and epigenetic clocks: a systematic review and meta-analysis of 140 studies

2026-06-12 · DOI 10.1038/S41562-026-02477-6

Y. E. Willems; A. D. Rezaki; M. Aikins; A. Bahl; Q. Wu; D. W. Belsky; L. Raffington

Abstract. Abstract Social determinants of health-such as socioeconomic status (SES) and race and ethnicity-strongly shape health. Individuals with lower SES or marginalized identities experience earlier onset of disease and shorter lifespans. 'Epigenetic clocks' measure biological ageing and are increasingly used to study healthy ageing, yet it remains unclear which clocks are most sensitive to social inequality. Here we conducted an Open Science Framework-pre-registered systematic review and meta-analysis of 140 studies (N = 65,919; 1,065 effect sizes) testing associations of SES and race and ethnicity with three generations of clocks. PubMed, PsycINFO, Web of Science, medRxiv and bioRxiv were searched from February 2024 to September 2025. Eligible studies included empirical articles in English published since 2013 on non-clinical populations reporting at least one relevant association. Associations between SES and biological ageing varied across different generations of epigenetic clocks ($F(2) = 178.10$, $P = -0.03$, 95% confidence interval (CI) [0.04, 0.01], $P = -0.11$, 95% CI [0.12, 0.09], $P = -0.13$, 95% CI [0.15, 0.11], P

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Decriminalizing same-sex relationships and its impact on social media discussions in Asia

2026-06-10 · DOI 10.1038/S41562-026-02485-6

Kokil Jaidka; Preetika Verma; Yiting Chen

Abstract. Abstract While legal reforms may coincide with shifts in public discourse, broader normative change often remains uneven and context dependent. Since 2018, four Asian countries-India, Taiwan, Thailand and Singapore-have decriminalized same-sex relationships, providing an opportunity to examine how such institutional changes are associated with public discussions of LGBTQ+ topics. Here, using supervised and unsupervised natural language processing methods, we analysed 136,255 Facebook news posts and 65,969 public Instagram captions within a 100-day window before and after each reform. To contextualize online discourse with offline sentiment, we incorporated nationally representative survey data from the Gallup World Poll (N = 46,200). Following decriminalization, online discussions were characterized by reduced moralized and affectively negative language, particularly in contexts with greater freedom of expression. By contrast, offline perceptions of neighbourhood safety for same-sex couples shifted more modestly and unevenly across age groups and countries.

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LGBTQ laws and public opinion change

2026-06-10 · DOI 10.1038/S41562-026-02424-5

Yu-Te Huang

Abstract. Abstract unavailable.

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A reporting checklist for large language models in behavioural science

2026-06-09 · DOI 10.1038/S41562-026-02492-7

Stefan Feuerriegel; Christopher Barrie; M. J. Crockett; Laura K. Globig; Killian L. McLoughlin; Dan-Mircea Mirea; Arthur Spirling; Diyi Yang; Tim Althoff; Maria Antoniak; Lisa P. Argyle; Ashwini Ashokkumar; et al. (81 total)

Abstract. Abstract unavailable.

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Proceedings of the National Academy of Sciences

Conceptual priorities shape individual gaze patterns during naturalistic visual attention

2026-06-12 · DOI 10.1073/PNAS.2604369123

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Abstract. Our visual landscape consists of not only people, places, and objects (e.g., "soldier," "stadium," "flag") but also the conceptual relationships that unite them (e.g., "patriotism"). Because conceptual knowledge varies across individuals, this level of structure may support individualized patterns of attentional selection during naturalistic scene viewing. Here, we ask whether individuals' gaze patterns reflect, in part, latent attentional priorities organized in conceptual space. Participants (N = 61) freely explored a diverse set of immersive real-world scenes (N = 100) in head-mounted VR while their gaze position was continuously recorded. We modeled gaze behavior using spatial, visual, and conceptual feature spaces, leveraging embeddings from large vision and language models, to uncover the latent priorities guiding individuals' unique patterns of selective attention across environments. Individuals exhibited stable and idiosyncratic gaze patterns across scenes and test-retest sessions, consistent with trait-like individual differences in attention. Spatial, visual, and conceptual feature spaces each explained unique variance in individual gaze patterns, with conceptual features contributing variance beyond that explained by spatial and visual features alone. Notably, language-model-based predictions were particularly effective at capturing these individualized patterns. Together, these findings indicate that naturalistic visual attention is structured at multiple levels-including a conceptual level-revealing stable individual differences in how people sample and prioritize information across complex visual environments.

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Misattributed biospheric carbon sink to the Three-North Program in the Taklamakan Desert may lead to an impractical policy: A commentary

2026-06-12 · DOI 10.1073/PNAS.2607916123

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Abstract. Abstract unavailable.

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Adaptive self-organization of global swidden forests

2026-06-10 · DOI 10.1073/PNAS.2606552123

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Abstract. Does swidden agriculture, a prototypical coupled human and natural system, exhibit a process of adaptive self-organization in which cultural practices balance environmental constraints through adaptive feedback? Here, we investigate whether quantitative signatures of adaptive self-organization can be detected in a dataset consisting of 18,000+ contiguous swidden patches in 18 remote sensing images of swidden mosaics from tropical and subtropical regions globally. We find that the distributions of patch sizes in 16 of 18 swidden areas exhibit power law patterns with scaling exponents 1, and correlation distances of 548 m. To account for these patterns, we develop a plausible ethnographically informed agent-based model of labor exchange, land use, and swidden site selection in which both sustainable and unsustainable

resource uses can emerge out of interactions among individuals or households. By analyzing the model, we identify spatial synchronization of swidden sites as the driver of power law formation, while social norms of swidden labor can guide the system to an intermediate level of landscape disturbance. Both mechanisms are required to maintain harvests and ecosystem productivity at high levels. Our model advances theoretical understanding of the socioecological dynamics of swidden agriculture, and supports the hypothesis that adaptive self-organization may be a general characteristic of coupled human and natural systems.

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Bridging national climate targets and firm-level decisions: An agent-based model of China's steel decarbonization pathways

2026-06-08 · DOI 10.1073/PNAS.2528009123

Xiao Liu; Jinhong Du; Lingling Lv; Jingnan Hu; Xingang Liu

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Abstract. China's steel sector is pivotal to national and global climate goals, yet decarbonization is constrained by entrenched coal-based processes and regional disparities. Top-down and bottom-up models often neglect firm-level heterogeneity and behavioral dynamics. Here, we develop an agent-based model linking policy signals-carbon caps, pollution standards, and resource limits-with firm-level investment decisions. We evaluate four scenarios combining current and strengthened climate and air targets with neutral or accelerated technology progress. All scenarios achieve deep CO₂ reductions (90 to 95% by 2060), but only the integrated policy-plus- technology pathway eliminates blast furnaces before midcentury, achieving the steepest declines. Policy- or technology-only pathways entail trade-offs, including lock-in to carbon capture and storage or delayed reductions. Our results show that synchronizing stringent climate policy with rapid technology cost declines is essential to steer China's steel industry toward a carbon-neutral trajectory.

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The Urban Pulse: Diagnosing the urbanization process as spiky, cyclical, and asynchronous

2026-06-08 · DOI 10.1073/PNAS.2537770123

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Abstract. Many urban indicators predominantly focus on static outcomes, failing to capture the dynamic nature of urbanization. Here, we introduce the concept of the Urban Pulse to characterize urbanization as a process of continual spatiotemporal change. Using dense time series remote sensing data, we operationalize the Urban Pulse for six cities that represent diverse development levels, economies, and geographies: Seattle, Shenzhen, Lagos, Mumbai, Dubai, and Mexico City. We then use the Urban Pulse to assess how COVID-19 affected each of the six cities. The results empirically substantiate three major insights. First, urbanization is inherently spiky, characterized by episodic events that deviate from baseline trends. Second, urbanization is cyclical, reflecting transitions through stages of expansion, stabilization, or decline, and does not occur with predictable frequency, such as seasonal patterns. Third, urbanization is asynchronous, with subcity areas exhibiting uncoordinated development patterns. The Urban Pulse COVID-19 analysis shows that while urbanization processes for some cities have not returned to prepandemic patterns, other cities exhibit a significant shift in patterns postpandemic. These findings suggest that the selection of a neighborhood or subcity area as an observation unit may have little bearing on a city as a whole. By capturing these spatial and temporal rhythms, the Urban Pulse framework provides a comprehensive understanding of process dynamics, enabling researchers to link processes to outcomes more effectively. This approach allows for the recognition of early warning signals at subcity scales and may facilitate targeted interventions at

the neighborhood level, where they are most impactful.

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Science

Patterns of brain-wide associations reflect socioeconomics

2026-06-11 · DOI 10.1126/SCIENCE.AEE6213

Scott Marek; Meghan Rose Donohue; Nicole R. Karcher; Caroline P. Hoyniak; Roselyne J. Chauvin; Ashley C. Meyer; John Miller; Andrew N. Van; Anxu Wang; Noah J. Baden; Vahdeta Suljic; Kristen M. Scheidter; et al. (36 total)

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Abstract. Previous brain-wide association studies (BWAS) have linked specific environmental and behavioral variables to brain variability. In this work, we mapped 649 variables to children's brains and compared the resultant BWAS maps with each other and with neurobiological reference patterns. Socioeconomic status (SES) showed the strongest brain-wide associations. The SES associations were strongest in motor and sensory but not cognitive regions, a pattern shared across many BWAS maps, including intelligence quotient (IQ). A single, common BWAS brain pattern existed across variables that was most reflective of a child's socioeconomics. Adjusting for SES weakened brain-IQ associations, eliminating the BWAS motor and sensory pattern. Brain-with-IQ associations also did not generalize when trained on higher-SES subsamples. Thus, children's brains vary the most with SES, potentially through SES-dependent sleep deprivation and stress.

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American Political Science Review

The Effects of Religious Messages and Endorsements on Political Attitudes: A Meta-Reanalysis

2026-06-10 · DOI 10.1017/S0003055426101695

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Abstract. The experimental study of the effects of religious messages and endorsements by religious leaders on political attitudes has a relatively short history yet has coalesced around three main claims: religious treatments are especially effective because of their religious character, effects are larger for religious affiliates than nonaffiliates, and effects are larger for high religiosity types than low religiosity types. Here, we meta-reanalyze the experimental record (58 treatment-outcome pairs drawn from 43 studies reported in 26 papers) to probe the generalizability of these claims. Our findings indicate that these three headline claims do not generalize straightforwardly across contexts: effects are large in some cases and close to zero in others, and we find no evidence in favor of the claimed heterogeneities by religious affiliation or religiosity. Based on a census of the estimands in this literature, we offer suggestions for future research that would enhance commensurability and synthesis.

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The Domestic Sources of International Trust

2026-06-08 · DOI 10.1017/S0003055426101713

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Abstract. How do states overcome mistrust? Scholars argue that costly foreign policy signals build trust. But when trust is low, such as during rivalries, states are unwilling to use these signals for fear of being cheated. We argue that domestic policies can also build trust by revealing information about a state's likelihood of cooperating internationally when there is a correlation between domestic and international preferences. We further argue that domestic policies have a distinct advantage: the value states accrue from them depends less on international reciprocation. As a result, domestic choices can reassure counterparts at moments when trust is so low that costly international signals appear prohibitively risky. We test the implications of our theory in case studies of the Cold War's end and United States-South Korea trust-building post-coup, illuminating several phenomena the current literature struggles to explain: initial trust-building between enduring rivals, asymmetric trust-building, and trust-building through illiberal domestic policies.

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The Expressive Duty to Vote

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Abstract. Standard arguments for the moral duty to vote in nationwide elections or referendums appeal to citizens' general duty of reciprocity not to free ride on the provision of the public goods of democratic governance and its benefits, to a nonreciprocal duty to maintain and strengthen democratic institutions, or to a general duty to bring about just outcomes. In this article, I argue that citizens are under an expressive duty to vote, as grounded in a more general duty to manifest support for just and democratic laws, and for democratic institutions as the instantiation of the requirement to treat one another with equal concern and respect. Even if (in circumstances I elucidate) citizens are morally permitted not to vote for either of the options on offer, it does not follow that they are morally permitted to abstain: sometimes, participating in the poll and voiding one's ballot is, on expressive grounds, the right thing to do.

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Nature

Deep learning four decades of human migration

2026-06-10 · DOI 10.1038/S41586-026-10611-7

Thomas Gaskin; Guy J. Abel

Abstract. Abstract Human migration is a fundamental driver of global demographic change, shaping population structure, labour markets and social policy across countries 1-3 . Although long-term migration patterns are often linked to economic development 4 , they can shift rapidly in response to shocks such as conflict, environmental crises and political change 5 . Despite its importance, migration remains difficult to measure consistently: existing data are sparse, concentrated in high-income settings and are fragmented across incompatible definitions, temporal resolutions and data types 6-8 . Past efforts have relied on partial datasets, including flow records, stock estimates and model-based reconstructions with limited coverage 9-14 . A central challenge is therefore to construct a globally consistent, high-resolution account of migration flows over time. Here we present a new dataset of annual origin-destination migration across 230 countries and regions from 1990 to the present, integrating diverse data sources into a unified modelling framework. By combining official statistics, census-based stocks, net migration estimates and past flow reconstructions, our approach produces temporally detailed and spatially comprehensive estimates that substantially extend existing resources. Using an ensemble of deep recurrent neural networks informed by geographic, economic, cultural and political covariates, we capture both persistent trends and short-term responses to changing conditions-all while propagating uncertainty to generate confidence bounds. Our results

outperform existing five-year flow estimates on held-out data and provide finer temporal resolution, revealing previously obscured dynamics in global migration patterns. This framework highlights regions in which uncertainty remains high and data collection is most urgently needed. By releasing all data, code and trained models, we provide a transparent and reproducible foundation for future work. These advances enable a more timely and detailed understanding of human mobility, with implications for research and policy in an increasingly dynamic global system.

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Political Communication

Evaluating Echo Chambers, Rabbit Holes, and Radicalization Pathways on YouTube

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Abstract. To what extent does the YouTube recommendation algorithm push users into echo chambers, rabbit holes, or radicalization pathways? Using a novel method to estimate the ideology of YouTube videos and an original audit design, we demonstrate that YouTube users are in mild ideological echo chambers, but this is primarily driven by user behavior. We also demonstrate that the recommendation algorithm prioritizes content that is similar to what the user is currently watching, producing content rabbit holes. However, we do not find evidence of radicalization pathways where users are driven into increasingly extreme content rabbit holes. Instead, we find that YouTube pushes all users, regardless of ideology, toward moderately conservative content and an increasingly narrow range of ideological content the longer they follow YouTube's recommendations.

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Multi-Platform Referrers of Misinformation: A Comparative Analysis of Misinformation Visits Referred by Facebook, Twitter, Instagram, Reddit, YouTube, Snapchat, and TikTok

2026-06-09 · DOI 10.1080/10584609.2026.2679492

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Abstract. Journalistic and scholarly accounts alike often depict social media platforms as key players in exposing people to misinformation. Unfortunately, existing research examining social media's role in exposing people to misinformation online implicitly operates under what we call the direct referrer paradigm, focusing only on visits to misinformation that directly follow social media. We introduce the embedded referrer paradigm, which considers social media's more comprehensive role in leading people to visit misinformation throughout their online journeys and operationalize it via counterfactual simulations. We estimate the effects of social media in exposing people to misinformation websites under both paradigms using 3 months of a large sample of Americans' (N = 1,240) web browsing behavior (21 million website visits). We explore seven different platforms' (Facebook, Instagram, Reddit, Snapchat, TikTok, Twitter, YouTube) effects in exposing people to misinformation sites with various ideological leanings. Overall, our results suggest that analyses under the direct referrer paradigm underestimate the role of social media in exposing people to misinformation, with estimated effects under the embedded referrer paradigm ranging from 1.5 to 15+ times larger than effects under the direct referrer paradigm. The difference in estimated effects under both paradigms is generally consistent across platforms, misinformation types (liberal, conservative, non-ideological), and individuals' partisanship. However, for some combinations of platforms, types of misinformation, and mobile vs. desktop devices, the discrepancy between the two paradigms is even greater.

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